PHAT-JeT: Patch Hierarchical Attention Transformer for Efficient Particle Jet Tagging

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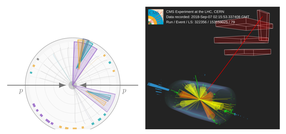
javierzhao.github.io/PHAT-JeT

1. Jet Tagging at the LHC Trigger

The LHC produces **40 million collisions per second**, a raw data stream above **1 PB/s**. Most events are background from known physics processes, and only a small fraction can be stored offline. The Level-1 trigger, running on FPGAs, decides within $O(10\ \mu s)$ which events to keep, while the budget for the jet tagging algorithm is **< 100 ns** (~1M FLOPs per jet).

A central trigger task is **jet tagging**: identifying the particle (quark, gluon, W, Z or top) that produced a jet, from up to 150 detected particles with no natural order.

Jets are particle sprays from short-lived decays; quark and gluon jets have one prong, W and Z two, top three. Small accuracy gains mean more rare signal events recorded.



Left: jets from a proton–proton collision. Right: CMS event display.

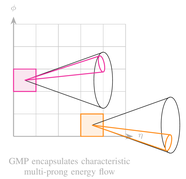
- No previous attention-based method achieves high accuracy and also fits within the strict trigger budget of **100 ns**.
- Low-rank, linear and hashing variants fit, but drop exact pairwise interactions and lose accuracy.

2. Related Methods

Method	Domain	Exact attention	Trigger-scale
Set Transformer	Sets	✗	✗
PaCa-ViT	Vision	✗	✗
Swin	Vision	✓	✗
ParT	Jets	✓	✗
JEDI-Linear	Jets	✗	✓
PHAT-JeT	**Jets**	✓	✓

- **Linformer:** low-rank projection compresses away particle–particle terms.
- **JEDI-Linear** (strongest FPGA baseline): linear-time, but no explicit pairwise edges.
- **HEPT:** hashing neighborhoods become unstable at small patch sizes.
- Swin and Longformer need a spatial grid or a meaningful sequence order; particle sets have neither.

3. PHAT-JeT Architecture

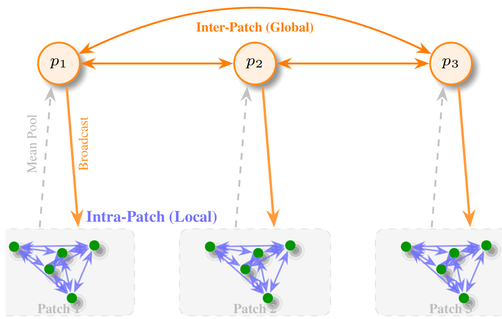


GMP encapsulates characteristic multi-prong energy flow

1 Geometric Message Passing

Each particle goes into a cell of a coarse (η, ϕ) grid of spacing δ. Features are summed per cell, a depthwise 2D convolution mixes neighboring cells, and the result is added back to each particle. Each particle then carries information about its angular neighborhood, capturing multi-prong structure.

$$G_{u,v} = \sum_{i:(u_i,v_i)=(u,v)} X_i, \qquad X_i \leftarrow X_i + W_p\,\mathrm{DWConv}(G)_{u_i,v_i}$$



2 Intra-patch (local)

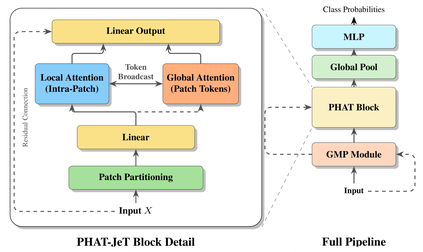
Particles are split into patches of P. Self-attention runs inside each patch j, so every pair in a patch interacts exactly. Cost $O(NP)$ instead of $O(N^2)$.

$$Z^{(j)} = \mathrm{Attn}\big(Q^{(j)}, K^{(j)}, V^{(j)}\big)$$
$$Q^{(j)} = X^{(j)} W_Q, \text{ etc.}$$

3 Inter-patch (global)

Each patch is averaged into a token p_j; tokens attend to each other, and each result is added back to the particles in its patch, so information flows across the whole jet.

$$p_j = \tfrac{1}{P}\textstyle\sum_i x_i^{(j)}$$
$$\tilde{p} = \mathrm{MHSA}(p_1, \ldots, p_{N/P})$$
$$x_i^{(j)} \leftarrow x_i^{(j)} + W_m\,\tilde{p}_j$$



Full model: input → GMP 1 → PHAT block (local 2 + global 3) → pool → MLP.

4. Datasets

hls4ml
CMS Level-1 trigger emulation; 5 classes.

JetClass
10 classes; trained on a 2M-jet subset.

Top Tagging
Top vs. light-quark/gluon jets.

Quark–Gluon
Quark vs. gluon jets.

Background rejection = 1 / (false-positive rate) at 80% signal efficiency: keeping 80% of signal jets, how many background jets are rejected for each one that passes. Higher is better; the main trigger metric.

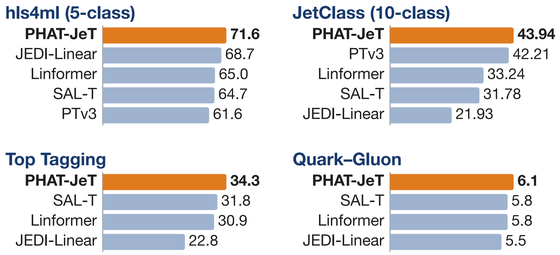
5. JetClass Results

128 particles per jet; efficient models at ~1.2–1.5M FLOPs.

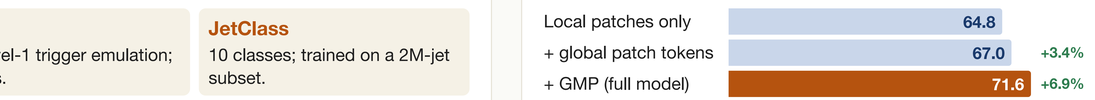
Model	Acc (%)	AUC	Bkg Rej	Params	FLOPs
PHAT-JeT	**65.38 ± 0.08**	**0.9391**	**43.94 ± 0.65**	**6.8K**	**1.24M**
Point Transformer V3	64.69 ± 0.11	0.9372	42.21 ± 0.25	23.7K	1.45M
Linformer	62.64 ± 0.19	0.9305	33.24 ± 0.11	9.5K	1.49M
SAL-T	62.19 ± 0.28	0.9291	31.78 ± 0.90	7.0K	1.42M
JEDI-Linear	59.52 ± 0.82	0.9192	21.93 ± 1.64	6.1K	1.31M
Small ParT	57.58 ± 0.28	0.9115	15.83 ± 0.43	2.9K	1.52M
Transformer	56.18 ± 1.34	0.9056	15.09 ± 2.53	1.0K	1.24M
PointNet	55.87 ± 0.16	0.9063	20.83 ± 0.70	9.9K	1.35M
Unconstrained references (10^2–10^4× more FLOPs)					
ParT (full)	*83.6*	*0.9834*	*2345.6*	*2,140K*	*340M*
L-GATr	*83.9*	*0.9842*	*2844.2*	*1,682K*	*9,709M*

PHAT-JeT vs. PTv3: **+0.69 pts** accuracy with 3.5× fewer parameters. vs. other efficient models: ≥2.7 pts.

6. Background Rejection by Benchmark



7. Ablation (hls4ml bkg rej)



Accuracy: 81.23 → 81.51 → 81.80%.

8. Beyond HEP: ScanObjectNN Results

Real 3D object scans (PB_T50_RS, 1024 points); same block, 3D voxel grid for GMP.

Method	Params (M)	OA (%)
PointNet++	1.5	77.9
SimpleView	0.8	80.5 ± 0.3
PHAT-JeT-XS	**0.157**	**81.22 ± 0.16**
PHAT-JeT-S	**0.612**	**82.29 ± 0.41**
PointMLP-elite	0.68	83.8 ± 0.6
PointNeXt-S	1.4	87.7 ± 0.4

Selected rows; baselines from published tables.

9. FPGA Estimates

AMD VU13P, 300 MHz, 64 particles; target **< 100 ns**; pre-route estimates.

Design	Acc	Bkg Rej	LUTs	ns
JEDI-Linear, official	80.94	66.0	69.9k	63.3
JEDI-Linear, official perm-invariant	81.81	80.4	158k	73.3
JEDI + GMP	82.08	84.8	144k	69.9
JEDI + GMP + patch tokens	81.93	84.0	208k	96.6
Patch attn (ReLU) + GMP	81.10	66.7	708k	99.9

Official JEDI-Linear designs use 64 particles; its performance degrades when extended to 128.

TL;DR: exact attention within patches and across patch tokens; best performance among trigger-scale models on four benchmarks.